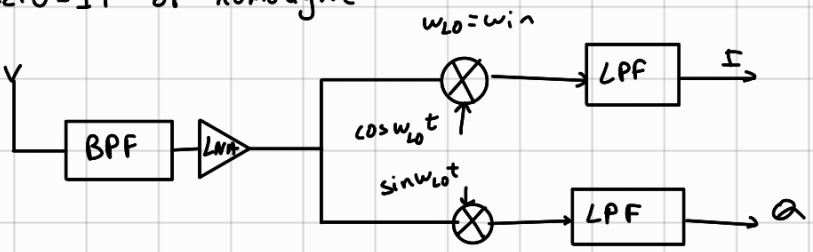
Direct Conversion Receiver

"Zero-IF" or "homodyne"



→ One step Conversion

Özellikleri

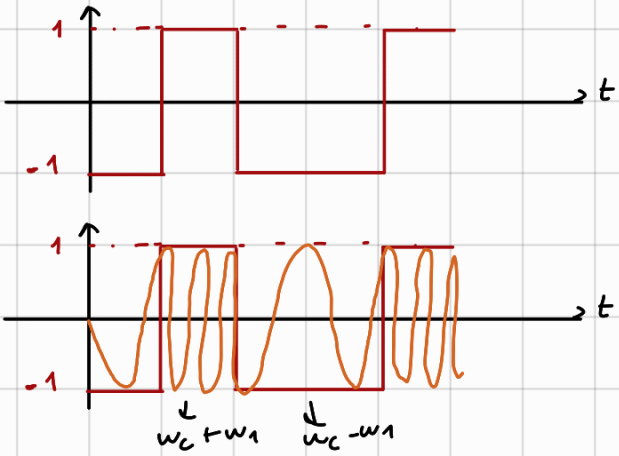
✓ No image

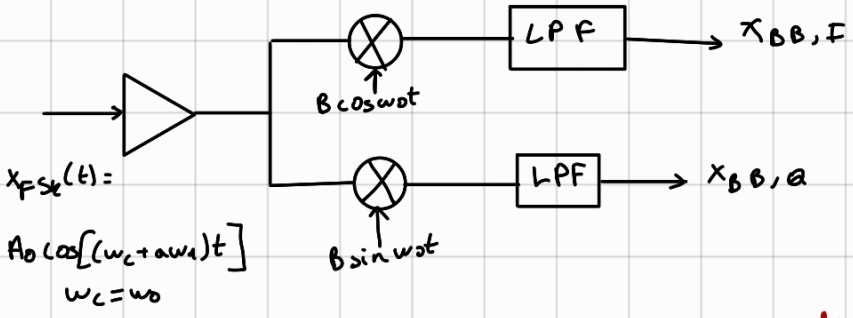
✓ Simple design process

✓ Mixing spurs are reduced

Ex: FSK Direct Conversion Receiver

$x_{FSK}(t) = A_0 \cos[(w_c + a w_1)t]$, $a = \pm 1$



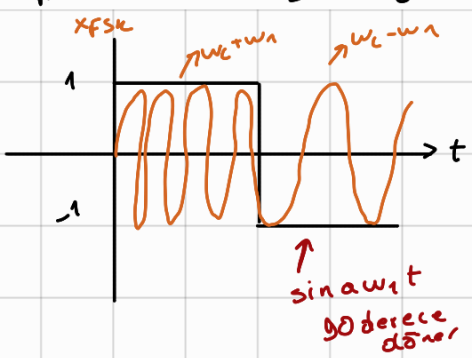


$$x_{BB,I} = A_0 \cos[(w_c + a w_1)t] \cdot B \cdot \cos w_0 t = \frac{A_0 B}{2}[\cos(2w_c + a w_1)t + \cos a w_1 t] \xrightarrow{LPF} x_{BB,I} = A_1 \cos a w_1 t$$

$$x_{BB,Q} = A_0 \cos[(w_c + a w_1)t] \cdot B \sin w_0 t = \frac{A_0 B}{2}[\sin(2w_c + a w_1)t - \sin a w_1 t]$$

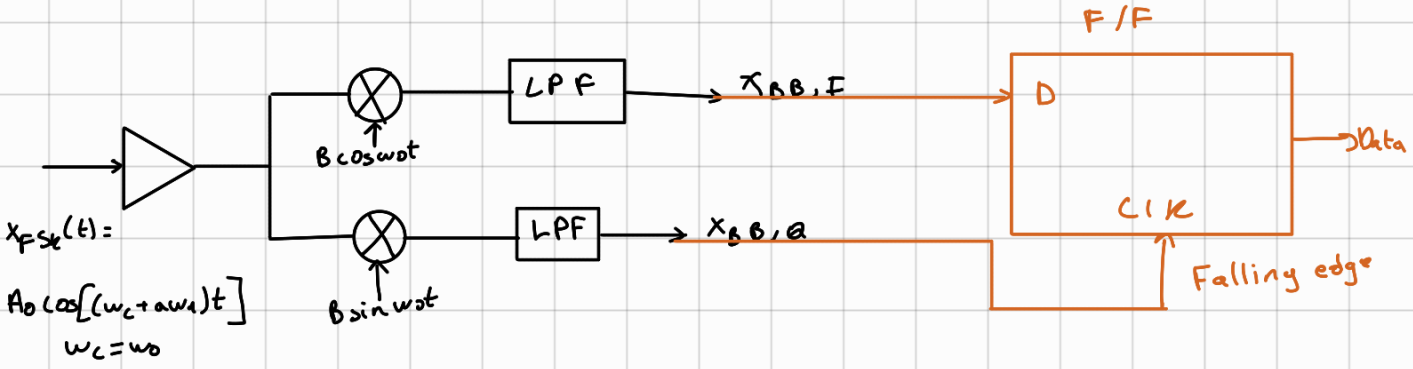
$$\hookrightarrow x_{BB,Q} = -A_1 \sin a w_1 t$$

Anlamak için 2 bit gönderdiğimizi düşünelim





Şimdi bu işlemi bir demodülatöre sokuyoruz

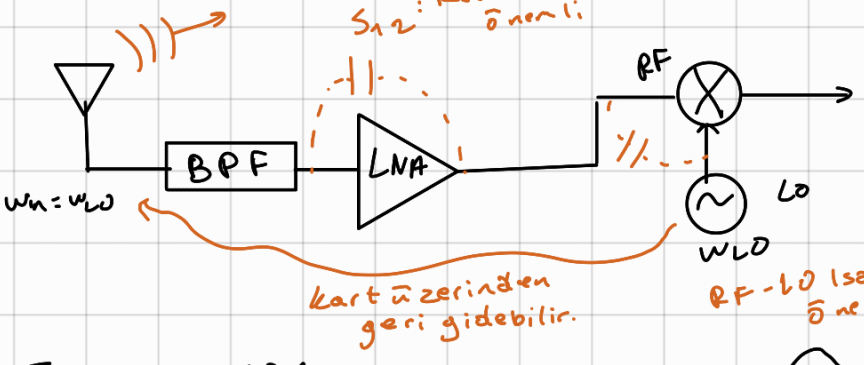


Drawbacks of Direct Conversion Receivers

① LO Leakage

önemli
S_{12}: Reverse Isolation önemli

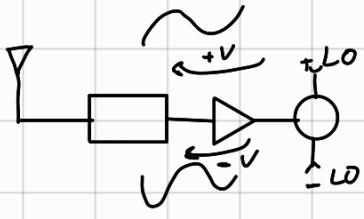
→ küçük olmasılazım $\log 0{,}000$ ← büyük değer



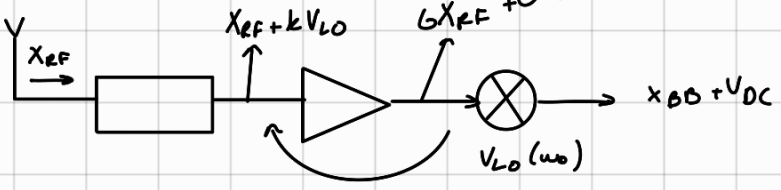
To minmize LO leakage

1) Differental Oscillator

2) Symmetric layout for Oscillator

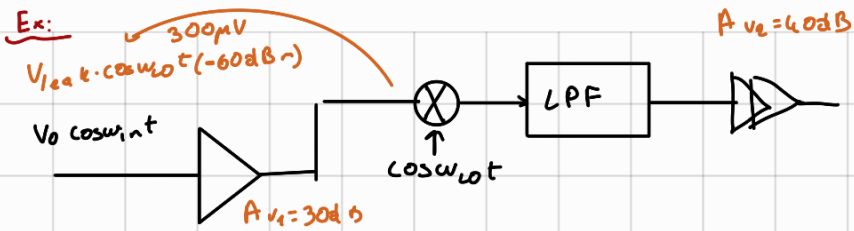


② DC offset



DC offsetten kurtulmalıyız

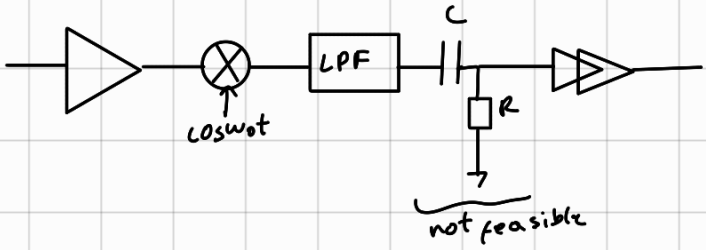
Ex:

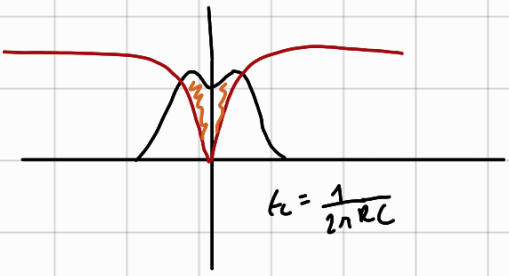


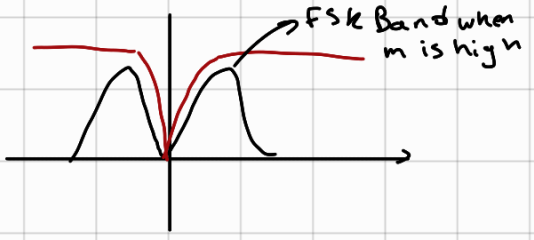
LO leakage at the input of LNA is -60 dBm

$-60+30+40 = 10$ dBm $\rightarrow V_{DC} \cong 1V$

Solving DC offset





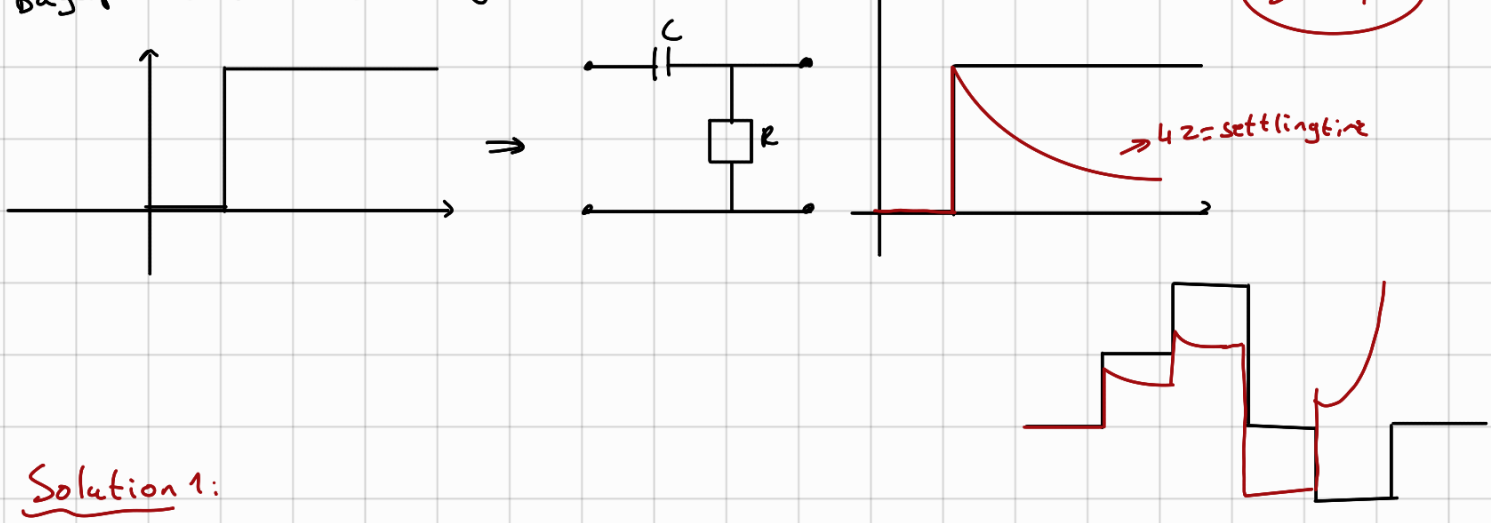


$m = \frac{f_2 - f_1}{f_c}$

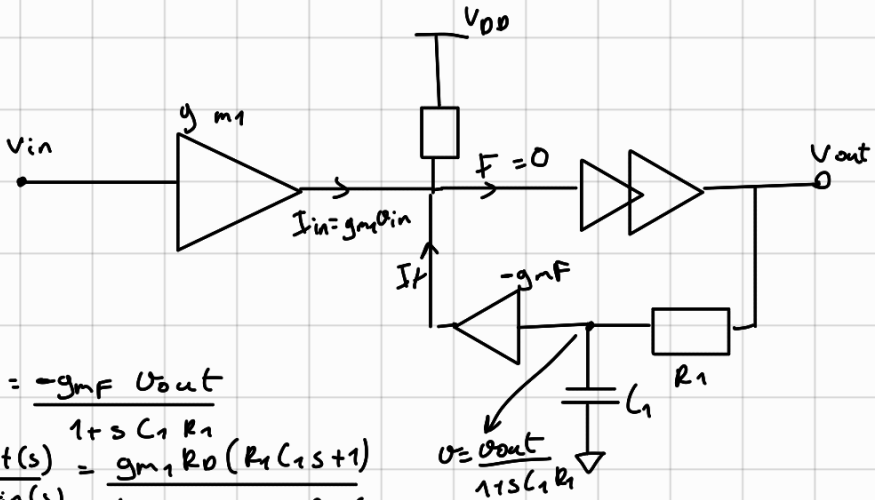
modulation index (FSK)

Bu gapının başka de zavantajı var

$Z = \frac{0.35}{f_c}$



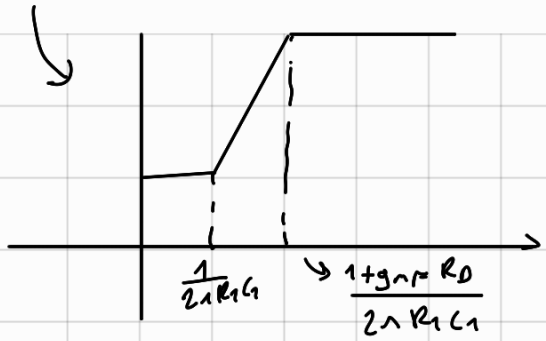
Solution 1:



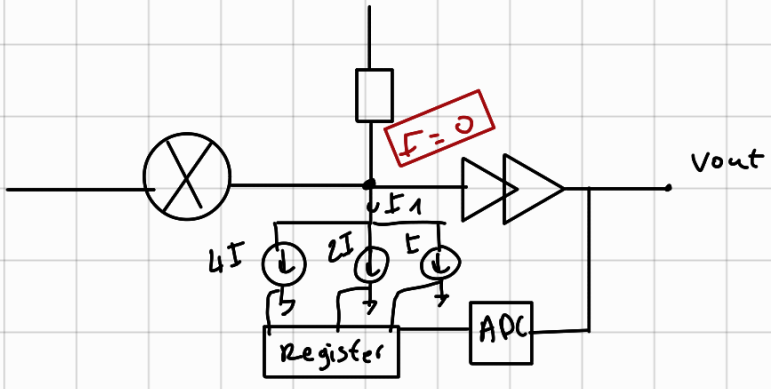
$$I_F = \frac{-g_{mF}\, v_{out}}{1+sC_1R_1}$$

$$\frac{v_{out}(s)}{v_{in}(s)} = \frac{g_{m1}R_0(R_1C_1s+1)}{1+g_{mF}R_0+sR_1C_1}$$

$v = \frac{v_{out}}{1+sC_1R_1}$

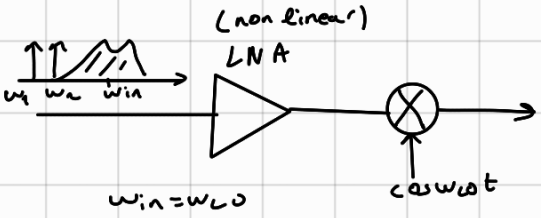


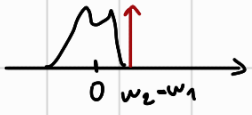
Solution ②



Drawback of Direct conversion Receivers

③ Even Order Distorsion





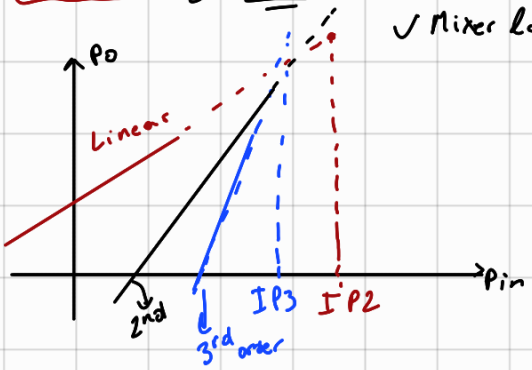
$2w_2 - w_1 \rightarrow 3^{rd}$ order distortion

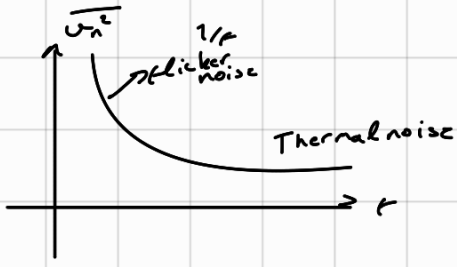
$w_2 - w_1 \rightarrow 2^{nd}$ (Even)

Solution ✓ High IIP2

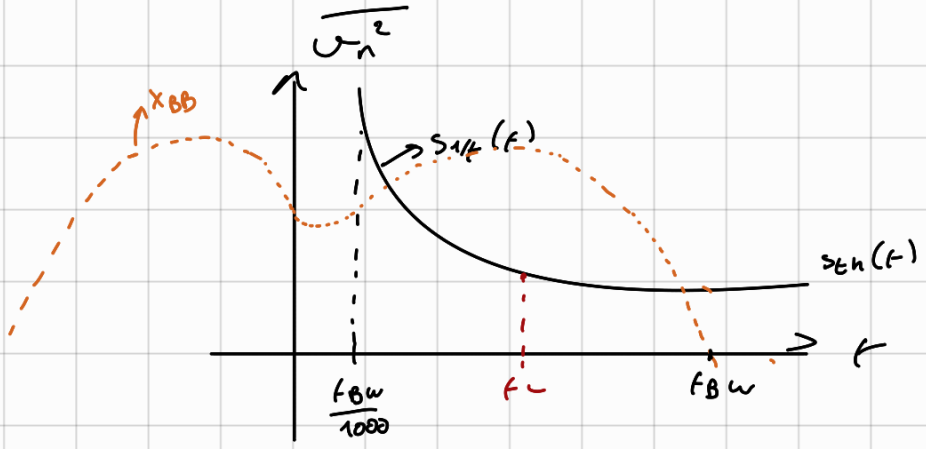
✓ Differential Mixer

✓ Mixer layout should be symmetrical





④ Flicker noise



f_{BW} : Half of RF channel Bandwidth

$S_{1/f}(f) = \frac{a}{f}$: PSD of flicker noise

$S_{th}(f) = S_{th}$: PSD of thermal noise

f_c : corner frequency

We ignore noise effect below $\frac{BW}{1000}$ since they vary slowly

$$S_{1/f}(f) = \frac{a}{f} \rightarrow S_{1/f}(f_c) = \frac{a}{f_c} = S_{th} \rightarrow a = f_c S_{th}$$

$$P_{n1} = \int_{\frac{f_{BW}}{1000}}^{f_c} \frac{a}{f}\, df + \int_{f_c}^{f_{BW}} S_{th}\, df$$

$$= \underbrace{a}_{\downarrow\, f_c S_{th}} \ln \frac{1000 f_c}{f_{BW}} + (f_{BW} - f_c) S_{th}$$

$$\boxed{P_{n1} = \left(5.9 + \ln \frac{f_c}{BW}\right) f_c S_{th} + f_{BW} \cdot S_{th}}$$

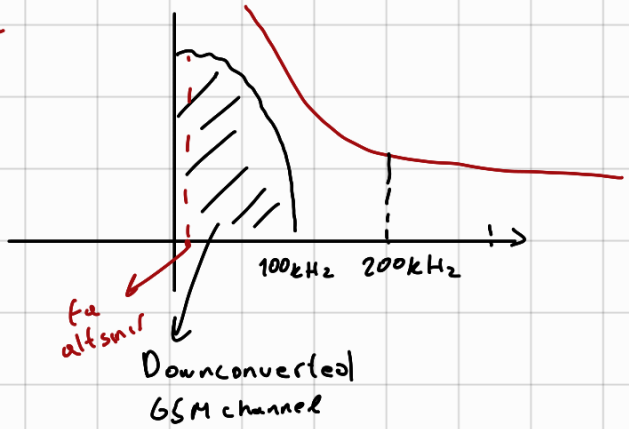
In the absence of flicker noise = $\boxed{P_{n2} \cong S_{th} \cdot f_{BW}}$

Flicker noise Penalty

↓↓ azaltmak istiyoruz

$$\boxed{\frac{P_{n1}}{P_{n2}} = 1 + \left(5.9 + \ln \frac{f_c}{f_{BW}}\right) \cdot \frac{f_c}{f_{BW}}}$$

Ex:



Flicker noise penalty

$$\rightarrow P_{n1} = \int_{f_{low}}^{100kHz} \frac{a}{f}\, df + \text{sonrasında işaret yok}$$

(normal dc fc ana bant bitiyor)

f_{low} çağrışımız $\frac{f_{low}}{1000}$ ⚠ formülü uygularken dikkat edin

$$= \int_{27Hz}^{100kHz} \frac{f_c S_{th}}{f}\, df = 8{,}2\, f_c\, S_{th}$$

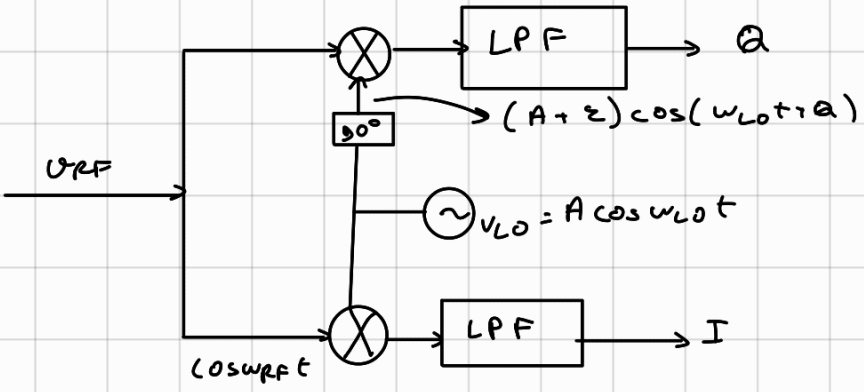
$100k$

$$P_{n2} = f_c S_{th} = 100k\, S_{th}$$

$100kHz$

$$\frac{P_{n1}}{P_{n2}} = \frac{8{,}2 \cdot f_c \cdot S_{th}}{S_{th} \cdot f_c} = 8{,}2$$

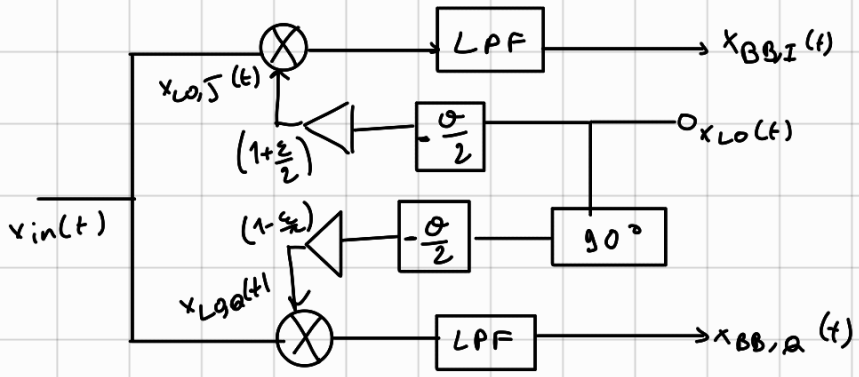
I/Q Mismatch (Drawbacks of Quadrature Downconversion) (up)



Phase & gain Error

$\varepsilon \rightarrow$ gain error

$\theta \rightarrow$ phase error



$x_{in}(t) = a\cos\omega_c t + b\sin\omega_c t \rightarrow$ QPSK signal

$x_{LO,I}(t) = 2\left(1+\frac{\varepsilon}{2}\right)\cos\left(\omega_c t + \frac{\theta}{2}\right)$

$x_{LO,Q}(t) = 2\left(1-\frac{\varepsilon}{2}\right)\sin\left(\omega_c t - \frac{\theta}{2}\right)$

When $\varepsilon = 0, \theta = 0$

$\rightarrow x_{BB,I}(t) = a$

$x_{BB,Q}(t) = b$

Outputs

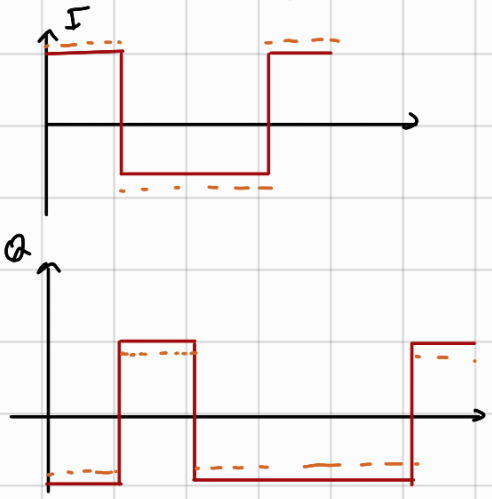
$x_{BB,I}(t) = a\left(1+\frac{\varepsilon}{2}\right)\cos\left(\frac{\theta}{2}\right) - b\left(1+\frac{\varepsilon}{2}\right)\sin\left(\frac{\theta}{2}\right) \longrightarrow$

$x_{BB,Q}(t) = -a\left(1-\frac{\varepsilon}{2}\right)\sin\left(\frac{\theta}{2}\right) + b\left(1-\frac{\varepsilon}{2}\right)\cos\left(\frac{\theta}{2}\right) \longrightarrow$

Only gain mismatch ($\theta = 0$)

$x_{BB,I}(t) = a\left(1+\frac{\varepsilon}{2}\right)$

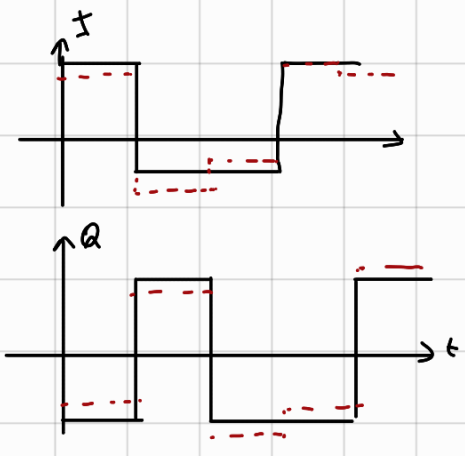
$x_{BB,Q}(t) = b\left(1-\frac{\varepsilon}{2}\right)$

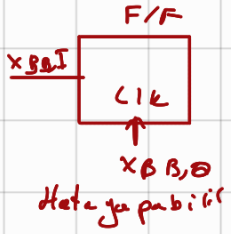


Only phase mismatch ($\varepsilon = 0$)

$x_{BB,I}(t) = a\cos\left(\frac{\theta}{2}\right) - b\sin\left(\frac{\theta}{2}\right)$

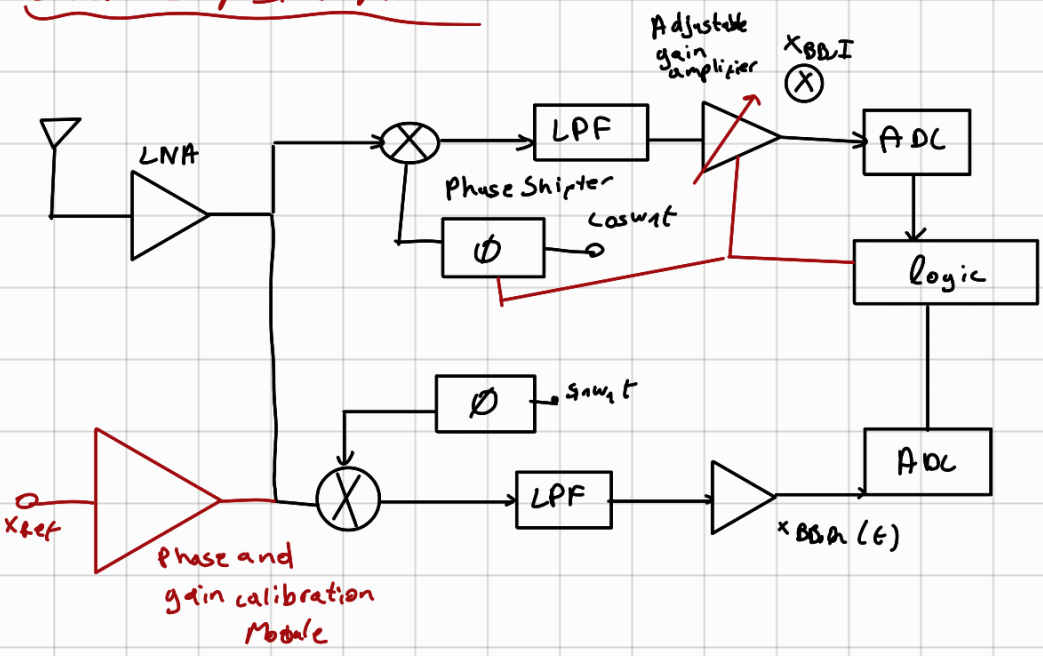
$x_{BB,Q}(t) = -a\sin\left(\frac{\theta}{2}\right) + b\cos\left(\frac{\theta}{2}\right)$





Bunu çözmek için ne yapmak gerekiyor

Solutions of I/Q Mismatch



I/Q Mismatch penalty